Where is the Price of Bitcoin Determined? Price Discovery in a Fragmented Market

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Abstract

This study examines the role of major Bitcoin exchanges in price discovery. Using high-frequency trade data from spot and perpetual futures markets involving fiat currencies and stablecoins, we identify spot and perpetual futures on the less-regulated exchange Binance as the primary source of price discovery. We show that lower transaction costs and higher trading volumes significantly enhance price discovery. Additionally, we analyze intraday dynamics and find that the more regulated spot exchanges, such as Coinbase, become relatively more influential during the New York 4 pm fixing. These findings highlight regulatory challenges, as Bitcoin prices used in products like ETFs are largely determined on exchanges susceptible to potential manipulation.

JEL Classification: C58, G1, G13, G14

Keywords: Bitcoin, Price discovery, Stablecoins, Perpetual futures

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We thank Jessica Gentner, Francesco Franzoni, Francesco Mazzari, Cara Stromeyer, and Alexander Wehrli for helpful comments and discussions. We thank Jan Gobeli for providing the data. The views, opinions, findings, and conclusions or recommendations expressed in this paper are strictly those of the authors. They do not necessarily reflect the views of the SNB. The SNB takes no responsibility for any errors or omissions in, or for the correctness of, the information contained in this paper.

1 Introduction

The rise of cryptocurrencies in recent years has created a new asset class that has captivated the interest of investors, regulators, and scholars alike. Bitcoin, the pioneer and dominant player among cryptocurrencies, has evolved into a significant global financial instrument since its inception in 2008 (Nakamoto, 2008) with a total market capitalisation surpassing 1700 billion US dollars (USD) and daily traded volumes ranging between 30 and 40 billion USD as of 2024.[1] However, the trading landscape is highly fragmented, with numerous exchanges offering a diverse range of products and operating under different levels of regulatory oversight compared to traditional financial markets. Many platforms provide high-leverage instruments such as perpetual futures and derivative contracts without expiration, allowing traders to hold positions indefinitely with high leverage. They are the most traded Bitcoin derivative, with volumes far exceeding the spot market. In addition, a significant share of Bitcoin trading occurs against stablecoins, digital assets pegged to fiat currencies that facilitate fund transfers while often bypassing regulation. The lack of regulatory oversight, coupled with the collapse of several major exchanges, has raised concerns about investor protection and market integrity in the Bitcoin ecosystem (Gensler, 2024). The problems of unregulated exchanges are further exemplified by the collapse of the exchange FTX in 2022, which highlighted the problems of little regulation in the industry.[2]

Price discovery, the process by which market prices incorporate new information, is a critical aspect of financial markets, if not the most important. This paper investigates the role of major cryptocurrency exchanges for Bitcoin price discovery. We pay particular attention to those exchanges used as benchmarks for pricing various financial products, such as exchange-traded funds (ETFs) and futures in traditional financial markets. In light of concerns regulators raise regarding the risks associated with unregulated trading platforms, we assess how these platforms shape price formation in the broader Bitcoin

[1] According to coinmarketcap.com.

[2] See: https://www.nbcnews.com/tech/crypto/ftx-freefall-was-oversight-rcna57098.

market. Specifically, we explore whether exchanges that offer Bitcoin trading against stablecoins play a significant role in determining Bitcoin prices on other platforms, potentially undermining the regulatory oversight in the Bitcoin market. Some exchanges, such as Coinbase, Bitstamp, or Kraken, are registered with regulators and comply with regulations (capital market regulations, client money custody regulations, know-your-client regulations, and anti-money-laundering regulations). However, some of the biggest exchanges, such as Binance, OKX, or Kucoin, operate outside the classical regulatory framework, and often, not even identity verification is required to trade on their platform. In our study, we define the registered, compliant exchanges as regulated, while those operating with minimal or no regulatory oversight are defined as unregulated.

This analysis utilises a high-frequency dataset encompassing trades across various exchanges and instruments that allow trading of Bitcoin against the USD and its stablecoin equivalent, Tether (USDT). To the best of our knowledge, this is the first study that includes the USDT market, a market that is five times bigger than the USD market,[3] to determine price discovery. Following a novel approach of Hasbrouck (2021), we leverage our high-frequency trade data with millisecond precision to identify price discovery within this fragmented market precisely. We provide estimates of the most common price discovery measures, including the Hasbrouck Information Share (HIS, Hasbrouck (1995)), the Component Share (CS, Gonzalo and Granger (1995); Harris et al. (2002)), and the Information Leadership Share (ILS, Putniņš (2013)). By employing multiple measures, we address the concerns highlighted by Dimpfl and Peter (2021), who pointed out that some price discovery metrics can be biased due to varying levels of market microstructure noise across instruments and exchanges.

Our analysis establishes a cointegration relationship between Bitcoin prices in USD and USDT markets to identify where prices are determined. By capturing the cointegration between different trading instruments, including stablecoin and fiat currency markets, we cover the largest share of trading activity in this fragmented market studied

[3]See: theblock.co.

to date. With our model and the high-frequency data used in this paper, we can precisely identify price discovery, addressing the limitations of previous research.

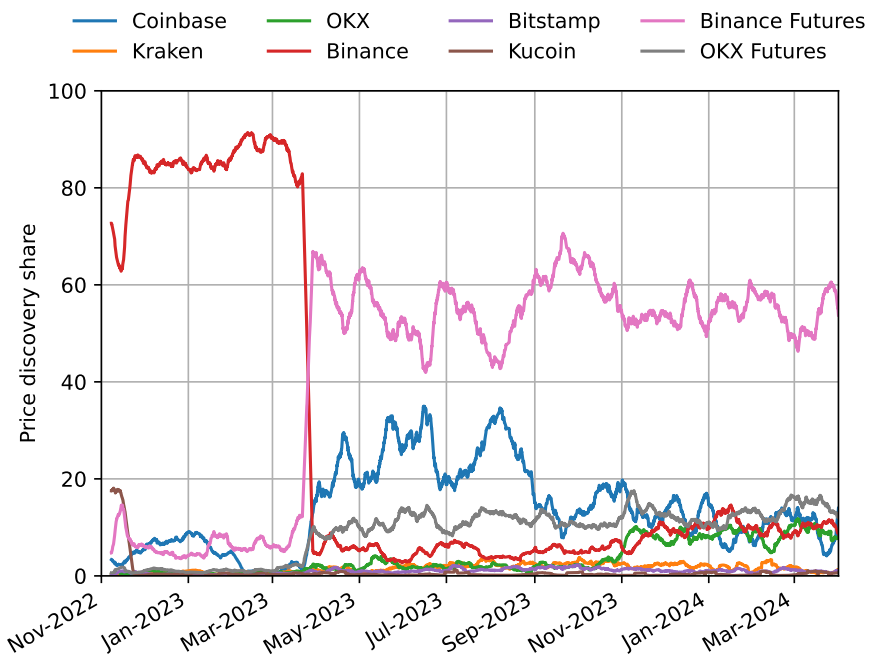

Figure 1: This figure shows the 7-day average of the price discovery share computed based on the information leadership share. The model includes traded prices from eight different exchanges and instruments between November 1, 2022, and March 31, 2024. The sharp drop coincides with March 22, 2023, when Binance ended zero-fee trading.

Our main results, displayed in Figure 1, reveal that spot and perpetual futures trades on Binance, the largest unregulated exchange, were the primary drivers of price discovery during our 17-month sample period from 2022 to 2024. These findings not only confirm that price discovery occurs in the most actively traded instruments but also highlight the significant influence of transaction costs. During part of our sample, Binance's marketing scheme allowed spot BTC trading against USDT without trading fees. The introduction of fees moved price discovery to the perpetual futures market, which then became the market with the lowest trading costs.

To further understand the importance of trading volumes and transaction costs, we provide a comprehensive empirical analysis of the dynamics of price discovery in the Bitcoin market. Our findings align with economic theory, showing that higher trading volumes lead to more informative prices (Admati and Pfleiderer, 1988). By estimat-

ing effective spreads and analysing a period of zero-fee trading, we can show that trading costs substantially impact price discovery. Furthermore, we document that higher realised volatility on unregulated markets correlates with increased price discovery, a phenomenon likely driven by more aggressive speculation (Alexander and Heck, 2020). Low trading costs, easy access, and a liquid and highly traded market make unregulated exchanges attractive for informed investors. Nevertheless, despite their relatively low trading volumes, USD spot markets, particularly Coinbase, play a significant role in price discovery.

To investigate the difference in price discovery among venues, we also study the intrady dynamics of price discovery and find that USD spot exchanges that contribute to the New York 4 pm fixing[4] are more important for price discovery during the fixing time window. This is important because this reference rate is the main benchmark for pricing other Bitcoin instruments, such as ETFs or regulated futures. Using the daylight change in New York as a quasi-natural experiment, we can establish a causal link between the fixing window and price discovery, similar to Krohn et al. (2024).

The fact that price discovery is dominated by an exchange that faces allegations of fraud and price manipulation (Crenshaw, 2024) presents significant challenges for regulators. This is particularly concerning as prices from such exchanges drive prices on other exchanges that serve as a benchmark for regulated Bitcoin products, such as ETFs or traditional futures, which continue to gain traction within the traditional financial sector.[5]

Our study contributes to several strands of the literature. We contribute to the field of research on price discovery in financial markets, which has been explored across various asset classes (Brogaard et al., 2014; Hauptfleisch et al., 2016; Hendershott et al., 2020; Chaboud et al., 2021). In the context of Bitcoin, Brandvold et al. (2015) and Pagnottoni and Dimpfl (2019) were among the first to analyze price discovery. The introduction of regulated Bitcoin futures by the CME has been a focal point in recent studies. While

[4]The CF Bitcoin Reference Rate - New York Variant (BRRNY) is calculated between 3 pm and 4 pm New York time, see https://www.cfbenchmarks.com/data/indices/BRRNY.

[5]See: https://www.bloomberg.com/news/articles/2024-12-30/blackrock-s-bitcoin-fund-became-greatest-launch-in-etf-history

earlier research suggested that spot markets dominated price discovery (Corbet et al., 2018; Baur and Dimpfl, 2019), later works confirmed the increasing influence of traditional futures (Kapar and Olmo, 2019; Hu et al., 2020; Fassas et al., 2020; Entrop et al., 2020; Aleti and Mizrach, 2021; Frino et al., 2025). Augustin et al. (2023) show that CME futures enhance market quality and price synchronization, whereas Alexander et al. (2020) and De Blasis and Webb (2022) emphasize the critical role of perpetual contracts. A key reference for our study is Alexander and Heck (2020), who analyze the interplay between regulated and unregulated markets. They find that CME futures play a limited role in price discovery, with unregulated spot and derivatives markets exerting greater influence. Makarov and Schoar (2019) find that USDT-based exchanges often lead price discovery. Our study extends the previous literature by integrating spot and perpetual futures from the most actively traded Bitcoin exchanges in USD and USDT into one multivariate model.

Additionally, we contribute to the literature on market quality and liquidity in Bitcoin spot and futures markets (Brauneis et al., 2021; Bianchi et al., 2022; De Blasis and Webb, 2022). We find that trading costs strongly influence price discovery, similarly to Albers et al. (2021). Moreover, we show that USDT spot markets can lead price discovery with zero transaction fees. Additionally, our results confirm that higher trading volumes enhance price informativeness, aligning with the theory of Admati and Pfleiderer (1988), and that increased realized volatility is associated with greater price discovery on certain exchanges.

Our study is the first to examine the intraday dynamics of price discovery and the significance of the New York Bitcoin fixing, linking it to the literature on foreign exchange fixing (see, e.g., Melvin and Prins, 2015; Evans, 2018; Krohn et al., 2024). We find that exchanges contributing to the fixing play a greater role in price discovery during that window. However, USDT-based exchanges continue to dominate price discovery, making them a crucial component of the fixing price.

Since potentially problematic exchanges dominate price discovery, concerns over market manipulation become increasingly relevant. Prior research has documented manipu-

lation tactics such as pump-and-dump schemes (Gandal et al., 2018; Li et al., 2021) and wash trading (Le Pennec et al., 2021; Aloosh and Li, 2024; Cong et al., 2023), while Griffin and Shams (2020) suggest that USDT issuance may influence Bitcoin prices. Our findings show that the prices of these products are primarily shaped by platforms facilitating Bitcoin trading through stablecoins and perpetual futures. Given the limited regulatory oversight in these markets, our results highlight important challenges for financial regulators.

The paper is structured as follows: Section 2 gives an overview of the institutional details of the Bitcoin market, describes the data we use, and presents descriptive statistics. Section 3 outlines different measures of price discovery and how we estimate them in a high-frequency setting. Section 4 presents and discusses the empirical results, while Section 5 concludes.

2 Institutional Details and Data

This section provides an overview of the Bitcoin market and the high-frequency data utilised in this study. We outline the market structure for Bitcoin trading, focusing on centralised exchanges, many of which operate with less regulation. We describe the data sources, exchanges, and instruments used. Additionally, we calculate market quality statistics, including effective spreads derived from trade data, and evaluate the performance of different methodologies. The section also presents summary statistics for the data used in this study.

2.1 The Market for Bitcoin

The rise of cryptocurrencies in recent years has created a new asset class, garnering attention from investors, regulators, and academia. Bitcoin stands out as the pioneer and dominant player among all cryptocurrencies. In 2023, almost 50% of all trading

in cryptocurrencies involved Bitcoin.[6] The popularity is shown by the fact that most instruments from the traditional financial sector, such as futures or ETFs, have been primarily designed to track the value of Bitcoin. Figure 2 shows the daily spot trading volume in Bitcoin and the Bitcoin price versus the US dollar over time.

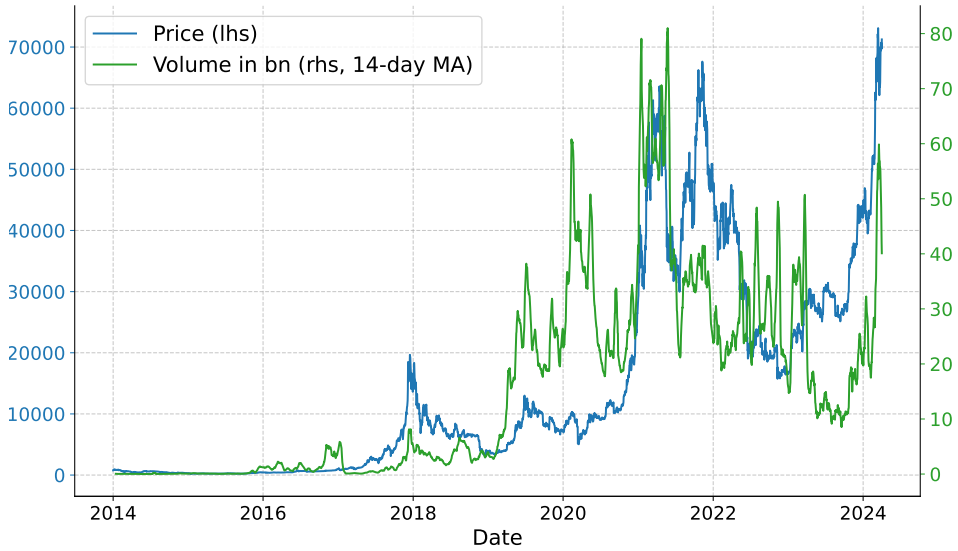

Figure 2: This figure shows the Bitcoin price in US dollars on the left-hand axis and the daily traded spot volume in billion US dollars on the right-hand axis. A 14-day moving average smoothes the trading volume, and the data is sourced from coingecko.com.

Numerous exchanges offer Bitcoin trading services, with centralised exchanges dominating the landscape. These exchanges function like traditional financial markets, utilising electronic limit order books to match orders centrally. Despite this, the market remains highly fragmented, with many platforms facilitating Bitcoin trading (coinmarketcap.com lists 239 different exchanges for trading Bitcoin). Centralised exchanges are more liquid and can swiftly incorporate new information into prices compared to their decentralised counterparts. Consequently, prices from these exchanges serve as a benchmark for determining the value of other financial instruments, such as futures or ETFs.

Investing and trading in Bitcoin is becoming increasingly popular among investors. As a result, at the beginning of 2024, the SEC approved the listing and trading of ETFs

[6]See https://assets.coingecko.com/reports/2023/CoinGecko-2023-Annual-Crypto-Industry-Report.pdf.

that physically hold Bitcoins for the first time. However, despite the approval, the SEC points out various risks regarding investments in Bitcoin and Bitcoin-related products. Among other aspects, explicit reference is made to the fact that the broader crypto market, especially unregulated trading platforms, may pose risks of fraud and manipulation to investors (Gensler, 2024). While traditional exchanges, like the New York Stock Exchange, are regulated under various financial laws, most centralised crypto exchanges operate outside the scope of such regulations. Cong et al. (2023) gives an overview of the regulatory status of the market. Furthermore, many exchanges have moved their headquarters to places without effective regulation.[7] The biggest centralised exchange by trading volume, Binance, even operates without official headquarters.

Given the problematic regulatory status of many exchanges, they face difficulties maintaining relationships with (regulated) banks and, thus, accepting and transferring traditional fiat currencies, this is especially true for the US dollar. As a popular solution, unregulated exchanges mainly offer trading against stablecoins. This also facilitates trading and transferring of funds. Stablecoins are cryptocurrencies pegged to another currency, commodity, or financial instrument. The most popular stablecoin by trading volume is USDT, which is pegged to the value of the USD and allows for dollar-like transactions without a banking connection. Given its popularity, much more Bitcoin is traded against USDT than the US dollar itself. Around 65% of total spot volumes are traded in USDT, compared to just 12% in USD, according to `theblock.co`.[8] Tether claims that the outstanding amount of USDT is backed by the same amount of US dollars held in assets by Tether. Owners can convert USDT into USD through exchanges or directly via Tether. While Tether remains the favoured currency for trading against Bitcoin, it has faced significant scrutiny. For instance, Griffin and Shams (2020) provide evidence of inadequate Tether reserves, proposing a hypothesis that Tether may be an unbacked

[7] In 2021, FTX relocated its headquarters from Hong Kong to The Bahamas, for a more favorable regulatory environment. See: https://www.bloomberg.com/news/articles/2021-09-24/bankman-fried-s-crypto-exchange-ftx-leaves-hong-kong-for-bahamas?embedded-checkout=true.

[8] Their calculation is based on Binance, Poloniex, Bitfinex, Huobi, OKX, Bittrex, Coinbase, Kraken, and Bitstamp volumes. The remaining amount has mainly been traded against other stablecoins.

digital currency inflating cryptocurrency valuations.

In addition to spot trading, Bitcoin futures have become a popular financial product, attracting significant trading volumes. Notably, perpetual futures contracts offered by many exchanges have attracted substantial demand. Initially proposed by Shiller (1993), perpetual futures are unique in that they lack an expiry date. To maintain the price close to the underlying spot rate, exchanges require participants to exchange cash payments, known as the funding rate, one to three times daily. Perpetual futures offer several advantages. Like spot products, they are straightforward to trade and do not involve rollover risk. Similar to traditional futures contracts, they also enable trading with high leverage. Consequently, trading volumes in perpetual futures markets on unregulated exchanges surpass those in spot markets. The detailed functioning of these contracts is well documented by Ackerer et al. (2023) and He et al. (2022).

2.2 Data

For our analysis, we utilise tick-level data from Tardis (see Tardis.dev), a provider of high-frequency order book and trade data from various cryptocurrency exchanges. Major cryptocurrency exchanges offer APIs for trading, and Tardis collects this data in real-time. The dataset, spanning from November 2022 to March 2024, is available with millisecond precision for all exchanges, and some exchanges even offer data at microsecond precision.

We selected the three most popular spot exchanges that offer trading against the USD: Coinbase, Bitstamp, and Kraken. These three exchanges are also the primary constituents of the widely-used CME CF Bitcoin Reference Rate, which determines the price of Bitcoin for many conventional financial products. For example, the iShares and ARK 21Shares Bitcoin ETFs and the CME futures contracts use this index as their benchmark. An oversight committee, including members from the CME, ensures that these venues adhere to legal standards and regulations and prevent illegal, unfair, or manipulative trading practices. Collectively, these three exchanges account for approximately 90% of

the trading volume of all exchanges contributing to the CME CF Bitcoin Reference Rate.[9]

For the mostly unregulated exchanges, we select three exchanges based on their reported trading volume. An obvious candidate to select is Binance, an exchange that, according to several sources[10] accounts for more than half of the global trading volume in Bitcoin. Moreover, we select OKX and Kucoin, other popular exchanges that lack formal regulation and report high trading volumes. Binance and OKX are both also dominant players in the perpetual futures market. Binance and OKX have a market share of around 60% of all traded perpetual Bitcoin futures contracts, according to theblock.co.

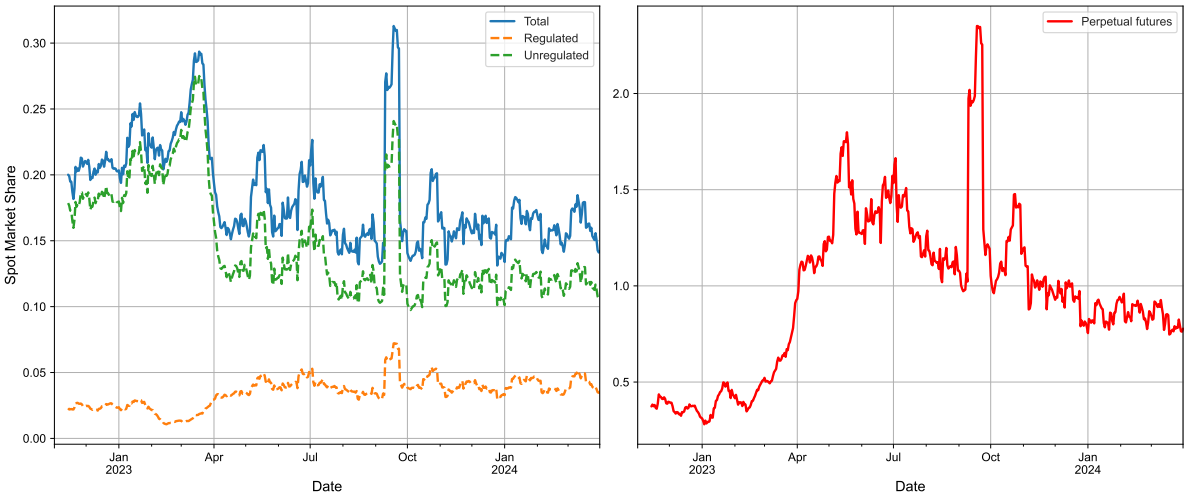

Figure 3: This figure shows the share of trading volume covered by our sample relative to the total Bitcoin spot market volume collected by coingecko.com. coingecko.com collects the Bitcoin spot trading volume for more than 200 exchanges from all around the world. The left-hand panel shows that our sample of spot trading volume consists of around 20% of the total trading volume. The right-hand panel shows the trading volume in perpetual Bitcoin futures on Binance and OKX relative to the spot market.

Figure 3 displays the trading volume in our sample relative to the total global Bitcoin spot market volume from over 200 exchanges. Our sample covers an essential part of the overall Bitcoin trading volume (around 20% of the total spot trading volume). Considering the highly fragmented market with numerous exchanges, most studies typically focus on a smaller subset of markets. To the best of our knowledge, this study represents

[9]The trading volume of the exchanges contributing to the CME CF Bitcoin Reference Rate and more details can be found under https://www.cfbenchmarks.com/data/indices/BRRNY.

[10]See for example, coinmarketcap.com, coingecko.com or theblock.co.

the largest market share by trading volume ever analysed. It also becomes clear that the unregulated exchanges have a much higher market share than the regulated ones. Moreover, the perpetual futures trading volume on Binance and OKX is much larger than the spot trading volume. Those two exchanges report trading volumes up to twice the daily traded spot volumes on over 200 exchanges.

Exchange	Instrument	Quote Currency	Tick size	Fees in bps (Taker/Maker)
Coinbase	Spot	USD	0.01	5 - 60 / 0 - 40
Bitstamp	Spot	USD	1	3 - 40 / 0 - 30
Kraken	Spot	USD	0.1	10 - 26 / 0 - 16
Binance	Spot	USDT	0.01	2.4 - 10 / 1.2 - 10 *
Binance	Futures	USDT	0.01	1.53 - 4.5 / 0 - 1.8
OKX	Spot	USDT	0.1	1.5 - 10 / -0.5 - 8
OKX	Futures	USDT	0.1	1.5 - 5 / -0.5 - 2
KuCoin	Spot	USDT	0.1	2.5 - 10 / -0.5 - 10

Table 1: This table shows the different exchanges and instruments used in this study. The exchanges charge different fees depending on whether a user adds liquidity (make) or removes liquidity (take). The fees typically vary depending on the user's average trading volume. All exchanges offer discounts for users who trade more frequently.
* Binance did not charge any fees for spot trading in BTC-USDT until March, 22 2023.

Table 1 provides an overview of the various exchanges and highlights their key differences. Notably, the tick sizes vary across exchanges, which can introduce different levels of microstructure noise and complicate the comparative analysis, as discussed by Dimpfl and Peter (2021). Additionally, the fee structures differ significantly among exchanges. For instance, KuCoin and OKX offer rebates for market makers, whereas regulated exchanges charge fees up to six times higher than the highest fees on unregulated exchanges. Moreover, Binance implemented an aggressive marketing strategy that included a fee waiver for BTC-USDT spot trading during part of our sample period. This campaign significantly boosted trading volumes on Binance, reinforcing its dominant market position. However, the end of this fee waiver scheme on March 22, 2023, resulted in a sharp decrease in spot trading volumes on Binance.[11]. Conversely, the decrease coincided with a

[11]See Galati (2024) for an analysis of the impact of the zero-fee program on market quality measures.

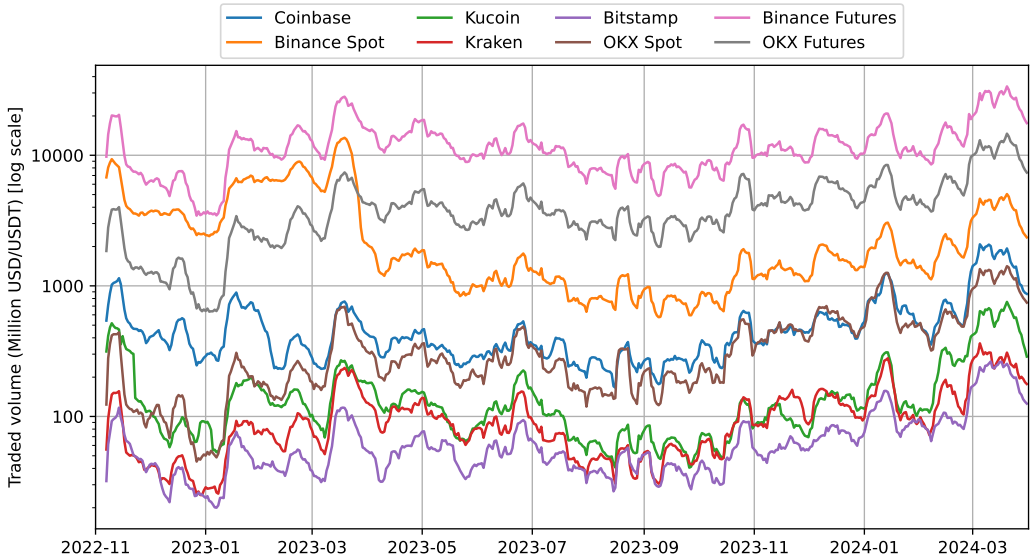

Figure 4: This figure shows the average daily traded volume, smoothened by the 7-day average, in million USD/USDT. The y-axis is depicted on a log scale.

rise in trading perpetual futures contracts on Binance. Since futures became relatively cheaper on Binance after the zero-fee program ended, many investors might have moved from spot to futures trading.

All exchanges offer trading 24/7, ensuring our data covers the entire sample period without interruptions. Figure 4 displays the average daily traded volume in million USD or USDT over time. The trading volumes across different exchanges and instruments generally move correlatedly. However, two notable observations stand out. First, the trading volume on Binance Spot dropped significantly after March 22, 2023, when the exchange introduced fees for trading Bitcoin against USDT. Second, trading volumes are relatively high at the beginning of the sample period and decrease towards the end. These trends are essential for understanding the dynamics of trading activity over the study period.

2.3 Transaction Costs

Beyond the explicit trading fees shown in Table 1, various implicit transaction costs can influence the attractiveness of an exchange for investors. To analyze the relationship between trading volume, transaction costs, and price discovery, we compute several transac-

tion cost measures. Specifically, in addition to trading volumes and returns, we calculate realised volatility, and effective spreads for all exchanges and instruments on an hourly basis. For realised volatility, we employ the two-scale estimator proposed by Zhang et al. (2005), which leverages our high-frequency data while mitigating microstructure noise that can distort estimates at very short time intervals.

Due to the lack of quote data, we estimate effective spreads using trade data for every individual market as prices do not have to align, given the fact that different instruments are involved. The effective spreads will later serve as a proxy for transaction costs on an exchange. Standard methods for estimating effective spreads include the Roll estimator (Roll, 1984), the Corwin-Schultz estimator (Corwin and Schultz, 2012), and the Abdi-Ranaldo estimator (Abdi and Ranaldo, 2017). However, Ardia et al. (2024) demonstrate that these estimators can be downward biased when trading is infrequent, and they propose an Efficient Discrete Generalised Estimator (EDGE) to address this issue. To determine the most accurate estimator for our data, we compare the performance of the Abdi-Ranaldo estimator with EDGE. We obtain one month of quote data for all exchanges and instruments, computing the hourly effective spread based on different sampling frequencies. The results in Table A.2 in the Appendix show that EDGE with a five-minute sampling frequency performs best for our case. Consequently, we use this specification to compute the effective spread.

Table 2 presents summary statistics for the different exchanges. As noted earlier, trading volumes on unregulated exchanges, particularly futures exchanges, exceed those on regulated exchanges. However, the volumes are highly variable on an hourly basis, as can be seen from the standard deviation. The average deal size is higher in futures trading due to the leverage available. All prices are quoted in USD or USDT, leading to closely aligned returns and volatilities across exchanges. Our estimate of the effective spread in basis points gives an indication of the transaction costs. USD-based exchanges have higher effective spreads, while Binance has very low transaction costs for spot and futures trading.

	Traded volume in million	Deal size in thousands	Return in %	Volatility in %	Spread in bps	Amihud in % per million
Coinbase Spot	21.27	1.25	0.72	27.85	1.26	1.91
	(30.59)	(0.72)	(49.59)	(27.67)	(1.13)	(1.82)
Kraken Spot	4.24	2.67	0.72	27.58	1.56	12.15
	(6.37)	(1.92)	(49.55)	(27.98)	(1.76)	(24.0)
Bitstamp Spot	2.8	3.18	0.69	27.62	2.83	17.39
	(4.6)	(1.93)	(49.46)	(27.4)	(2.34)	(18.47)
Kucoin Spot	6.56	0.6	0.7	27.88	1.0	6.31
	(10.6)	(0.4)	(49.3)	(27.76)	(1.07)	(5.57)
Binance Spot	114.44	1.03	0.7	27.83	0.36	0.4
	(146.42)	(0.34)	(49.28)	(27.66)	(0.71)	(0.38)
Binance Futures	511.15	3.17	0.66	27.88	0.33	0.07
	(638.71)	(0.75)	(49.45)	(28.04)	(0.42)	(0.05)
OKX Spot	15.22	1.33	0.7	27.9	1.1	3.51
	(25.7)	(0.53)	(49.41)	(28.19)	(1.02)	(3.97)
OKX Futures	173.63	5.54	0.68	27.94	0.72	0.24
	(235.94)	(1.22)	(49.45)	(28.24)	(0.85)	(0.23)

Table 2: Summary statistics based on hourly data from November 1, 2022, until March 31, 2024. We display the average value for trading volumes, deal size, returns, realized (annualized) volatility, the estimated effective spreads, and the Amihud (2002) measure. The standard deviation is depicted below in brackets.

3 Methodology

This section presents the model and the various price discovery measures employed in our analysis. It also details the sampling frequency and lag selection criteria used. The section concludes with the cointegration results for our different time series, emphasising the importance of cointegration as a prerequisite for estimating price discovery with our model.

3.1 Model

The model specification to estimate price discovery is as follows: Let $p_t = [p_{t,1}, p_{t,2}, ..., p_{t,n}]'$ be a vector of prices with covariance stationary first differences. These can be quoted or traded prices from different exchanges. They can also contain spot or perpetual fu-

tures prices and belong to different underlying cryptocurrency pairs (BTC-USD or BTC-USDT). Importantly, they all have to possess a common trend, i.e., this means in our setting that the prices have to be cointegrated of order $n-1$. In the following, we assume that this assumption holds, but we will show in Section 3.3 that our data is cointegrated of order $n-1$.

Following Engle and Granger (1987) and Hasbrouck (1995), we can write the dynamics as a vector error correction model (VECM), including lags up to k:

$$\Delta p_t = \gamma\beta p_{t-1} + \sum_{i=1}^{k} \Gamma_i \Delta p_{t-i} + \varepsilon_t, \tag{1}$$

where βp_{t-1} contains the $n-1$ cointegrating vectors that are $p_{t,1}-p_{t,2}, p_{t,1}-p_{t,3}, \ldots, p_{t,1}-p_{t,n}$ (i.e.,

$$\beta = \begin{bmatrix} 1 & \\ \vdots & -I_{n-1} \\ 1 & \end{bmatrix}$$

). γ is the $n \times (n-1)$ matrix of adjustment coefficients and $\varepsilon_t \sim (0, \Omega)$.[12] It is also important to note that perpetual futures can systematically deviate from the underlying spot price due to the funding payment (Ackerer et al., 2023). However, the funding payment on an hourly level is typically very small, often less than 0.01% of the position's notional value. Consequently, we follow the approach of previous studies and omit it from our model (Alexander et al., 2020; Alexander and Heck, 2020).

The VECM is covariance stationary and, therefore, admits the Wold representation:

$$\Delta p_t = \Psi(L)\varepsilon_t, \qquad \text{where } \Psi(L) = \sum_{s=0}^{\infty} \Psi_s L^s.$$

Using the Beveridge-Nelson decomposition (Beveridge and Nelson, 1981), i.e., we write $\Psi(L) = (1-L)\Psi^*(L)$, we can find the common trend representation:

[12]Note that Equation 1 does not contain an intercept. This can be done due to the fact that we demean all price series. This ensures that $\mathrm{E}(p_i - p_j) = 0$ for all time series $i \neq j$.

$$p_t = p_0 + \Psi(1)\sum_{s=0}^{t}\varepsilon_s + \Psi^*(L)\varepsilon_t, \qquad \text{where } \Psi(1) = \sum_{k=0}^{\infty}\Psi_k. \tag{2}$$

$\Psi(1)$ represents the overall effect of innovation ε_t on future price movements, indicating the long-run impact of ε_t on prices. Thus, the second term on the right-hand side of the equation above is the random walk component that drives the prices in the system, while the third term is the transitory component. Due to cointegration, all rows of $\Psi(1)$ are identical. Let ψ denote any row of $\Psi(1)$. The variance of the random walk component can then be written as:

$$\sigma_w^2 = \psi\Omega\psi'.$$

The information share of market i is now defined as the share of variance of the random walk component that is attributed to the innovations of market i:

$$HIS_i = \frac{\sigma^2_{\varepsilon_{t,i}}}{\sigma_w^2} = \frac{\psi_i^2\Omega_{i,i}}{\psi\Omega\psi'}, \tag{3}$$

where ψ_i is the i-th entry of ψ corresponding to market i. Ω will be diagonal if the error terms $\varepsilon_{t,i}$ are mutually uncorrelated. However, if the error terms are correlated, it becomes clear that the information shares are not uniquely defined. In such cases, Hasbrouck (1995) proposes that they can be characterised by upper and lower bounds defined by alternative Cholesky factorisation of Ω. One can write the error term in terms of orthogonal innovations u_t as:

$$\varepsilon_t = Fu_t \tag{4}$$

where F is the Cholesky decomposition $FF' = \Omega$ of Ω. Thus, it is possible to define the IS for the i-th market as:

$$HIS_i = \frac{([\psi' F]_i)^2}{\psi\Omega\psi'} \tag{5}$$

The upper and lower bounds then follow from the different causal ordering of the variables in the VECM. Therefore, they cannot be interpreted as confidence bounds. Moreover,

the upper and lower bounds can often be large and overlapping, a point that is criticised in empirical applications (see for example Yan and Zivot, 2010) as the method does not produce meaningful results.

Working with higher resolutions can help to overcome this problem. Decreasing the sampling interval reduces contemporaneous cross-correlations between different exchanges or instruments. By construction, this will lead to a much narrower IS bound that does not have the previously discussed identification problem. Nevertheless, higher resolutions come with computational costs as lags increase drastically. For example, one day of market data with a frequency of one second contains at most 86'400 entries. In comparison, one day in the highest available frequency (1 millisecond) can contain up to 86.4 million entries. Moreover, depending on the number of lags, Equation 1 will have millions of coefficients. To limit the number of coefficients of the VECM, Hasbrouck (2021) adapts the heterogeneous autoregressive (HAR) model of Corsi (2009) to a VECM setting to reduce the number of coefficients in the model.

We follow this approach and estimate the model at progressively higher resolutions. We define d as the resolution of one second for the frequencies of 1s, 100ms, 10ms, and 1ms, i.e., $d = 1$ for 1s and $d = 0.001$ for 1ms. In this case, Δp_t becomes $\Delta_d p_t = p_t - p_{t-d}$, and for simplicity, we follow Hasbrouck (2021) and set the maximum lag to 10s across all resolutions. Therefore, we can adjust the model in Equation 1 and estimate the following model:

$$\Delta_d p_{i,t} = \gamma\beta p_{t-d} + \sum_{j=1}^{n}\sum_{k=1}^{K} \Gamma_k^{ij} \Delta_d p_{j,t-kd} + \varepsilon_{it}, \tag{6}$$

where $K = 10/d$ and $t = d, 2d, \ldots, T$. For each pair of i and j, there are K autoregressive coefficients. They are then restricted as in the HAR model of Corsi (2009). This means that for a frequency of 1s, the coefficient of Γ_1 is unrestricted and $\Gamma_2 = \cdots = \Gamma_1 0$. For a frequency of 0.1s Γ_1 is unrestricted, $\Gamma_2 = \cdots = \Gamma_{10}$ and $\Gamma_{11} = \cdots = \Gamma_{100}$. For a frequency of 0.01s Γ_1 is unrestricted, $\Gamma_2 = \cdots = \Gamma_{10}$, $\Gamma_{11} = \cdots = \Gamma_{100}$ and $\Gamma_{101} = \cdots = \Gamma_{1000}$. As in Hasbrouck (2021), with every increase in resolution by a factor of 10, the nearest lag term is substituted by ten terms of higher resolution. The initial coefficient is unconstrained

among these higher-resolution terms, while the last nine are fixed to the same value. This reduces the number of coefficients drastically and makes it feasible to estimate the model. However, it is still costly to estimate the cross-product matrix of the VECM; nevertheless, by applying efficient calculations of cross-product matrices[13] estimation of the model with data in microsecond resolution becomes feasible. [14]

Furthermore, to serve as a benchmark to information shares as proposed by Hasbrouck (2021), we can use the model from Equation 6 to compute the Component Share (CS, Gonzalo and Granger, 1995; Harris et al., 2002), which is another measure of price discovery that is widely used in the literature. To compute the CS, we can use γ, the matrix of adjustment coefficients within the VECM. As described in Gonzalo and Granger (1995) and Baillie et al. (2002) for every vector γ_i in γ we can derive the orthogonal vectors $\gamma_{\perp,i}$. The component share for the i-th market is then defined as:

$$CS_i = \frac{\gamma_{\perp,i}}{\sum_{i=1}^{n} \gamma_{\perp,i}}. \tag{7}$$

As discussed and highlighted in Dimpfl and Peter (2021), comparing different exchanges involves a lot of microstructure noise, and it is possible that the HIS or the CS could be biased. Putniņš (2013) mentions several aspects that are relevant in our case, among others, different tick sizes and liquidity effects as microstructure noise. Moreover, Baur and Dimpfl (2019) show that this can be especially the case for cryptocurrencies. Yan and Zivot (2010) and Putniņš (2013) address this issue and develop the Information Leadership Share (ILS) that combines the HIS and the CS to cancel out potential noise. Based on their work, Patel et al. (2020) formulate the multivariate ILS for a VECM with more than two price series, which we will primarily use in this paper. The ILS is defined

[13] See https://pages.stern.nyu.edu/~jhasbrou/Research/HRVAR/HRVARindex.html.

[14] Moreover, we rewrite the code by Hasbrouck (2021) in Python to further improve computational efficiency and optimally use parallelised computing in a high-performance computing (HPC) cluster. The code is available from the authors upon request. It is important to note that even with this procedure, the amount of data at the millisecond level is still substantial.

as:

$$ILS_i = \frac{\beta_i^2}{\sum_{i=1}^{n} \beta_i^2}, \tag{8}$$

where $\beta_i = \frac{HIS_i}{CS_i}$. Given the different levels of microstructure noise across various platforms and instruments, we prioritise the ILS as the primary measure for determining price discovery. For comparative purposes, Figures A.2 and A.3 in the Appendix provide the price discovery share results measured by the HIS and the CS, respectively.

The main advantage of the model and the high-frequency data used in this paper is that we can identify price discovery as measured by the HIS. Figure 5 illustrates this by displaying the HIS for two sampling frequencies. Specifically, Figure 5 shows the HIS for OKX in November 2022 calculated with data at a one-second frequency and a ten-millisecond frequency. It becomes clear that with the one-second frequency, which is the highest frequency of the data used in the literature, the upper and lower bounds of the Hasbrouck (1995) model are extremely wide and cover almost the entire range between 0 and 1. Thus, identification of the information share is almost impossible. This is the main reason why many other studies of price discovery in the Bitcoin market do not use the HIS and focus exclusively on the CS. Different studies use the mean of the upper and lower bound of the HIS to approximate the actual value. These bounds are typically wide, and they are frequently overlapping in a setting with multiple time series. However, as shown here, using higher frequency data and the estimation method previously described reduces the bounds significantly and provides meaningful estimates that can be used to study price discovery.

Figure A.1 in the Appendix shows a simple model with three price series. In the picture, the high level of uncertainty in the HIS estimates at lower frequencies translates to imprecise estimates of the ILS. Using our proposed model can also resolve these issues and provide clear answers to where price discovery occurs.

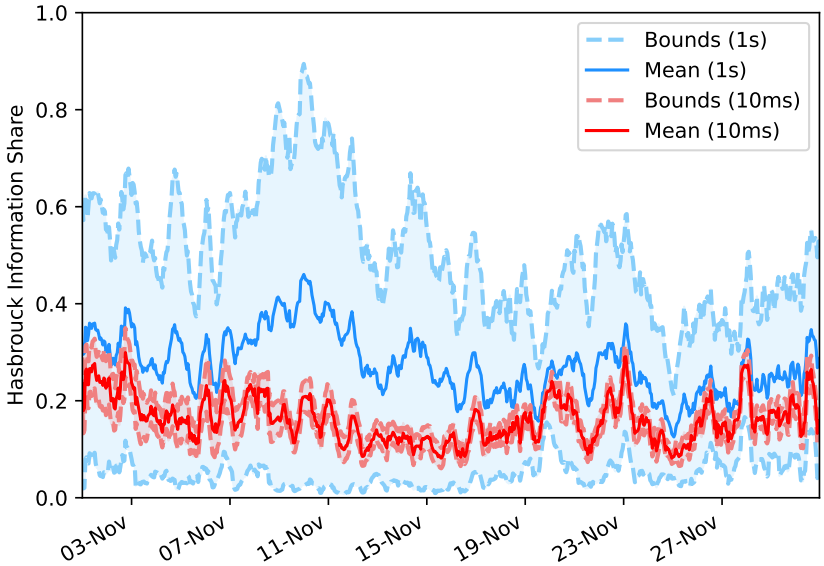

Figure 5: This figure shows the Hasbrouck Information Share (HIS) for trades on OKX in November 2022. It displays estimates of the HIS, including the upper and lower bounds for two models based on different sampling frequencies. The first model aggregates traded prices for every second (1s), and the second model provides information at a ten millisecond (10ms) level.

3.2 Model Specifications

We are interested in measuring daily and hourly effects (e.g., around the Bitcoin price fixing). Therefore, we estimate the model in an hourly frequency. First, we estimate a model with only spot prices from the three regulated and three unregulated exchanges. We aggregate the traded prices at different frequencies and start with the average traded price every second. Afterwards, we reduce the sampling frequency to 100 milliseconds, 10 milliseconds, and one milliseconds, which is the highest available frequency for some of the exchanges in our dataset. We follow the proposal of Hasbrouck (2021) and use the same polynomial lag structure as in the original paper (see Table 1 in Hasbrouck, 2021). We find that the upper and lower bounds of the HIS become narrow (i.e., the shares' upper and lower bounds no longer overlap and are identifiable) at a frequency of 10 milliseconds. Given that the computation becomes exponentially more expensive for

higher frequencies, we use the frequency of 10 milliseconds for the final application.[15] We use all available data and estimate one model for every hour between November 1, 2022 and March 31, 2024. We estimate the model for eight different time series at the same time. As the identification of the HIS is defined via permutation of all options, it becomes difficult to compute the measure for more than six time series in one model. Therefore, we removed the two spot price series with the lowest contribution to price discovery to analyse the futures markets and replaced them with the Binance and OKX futures series.

3.3 Cointegration

As previously mentioned, a prerequisite of the model adopted is that the time series must be cointegrated of order $n-1$. In this subsection, we report the cointegration results and show that the time series of Bitcoin prices are cointegrated on the different exchanges, most importantly, among USD and USDT. Here, we report the results of a cointegration analysis based on our baseline model that consists of hourly estimations with eight time series at a 10-millisecond frequency. We use the Johansen (1991) cointegration test to analyse the cointegration and show that the time series with spot and futures prices from different exchanges are cointegrated. Table A.1 in the Appendix shows, as an example, the results of the Johansen test for one hour of data. We utilise the Johansen methodology to compute both the trace and maximum eigenvalue test statistics at a 1% confidence level as in column Trace and column Lmax, respectively. If we can reject all the hypotheses that $r = 0, 1, 2, 3, 4, 5, 6$, we conclude that there must be at least seven cointegrating relationships within our eight time series and that the previously described method to estimate price discovery is valid. In our sample, this is the case for 87% (91%) of all hours at the 1% (5%) confidence level. We discard the observations where the cointegration relationship does not hold from our dataset to prevent potentially distorted results (Entrop et al., 2020).

[15] Results for other frequencies are available by request from the authors.

4 Empirical Results

This section presents the findings from our empirical analysis, focusing on the results of our price discovery measures over time. It explores how higher trading volumes and lower transaction costs are significantly associated with higher price discovery. Additionally, we provide insights from a quasi-natural experiment that examines the impact of the Bitcoin New York fixing on price discovery. The section continues with an analysis of the perpetual futures basis and the influence of macroeconomic releases on price discovery. We conclude with several robustness checks for the validity of the empirical results.

4.1 Price Discovery

In this subsection, we present the results of the price discovery share for our models. Figure 1 illustrates the price discovery shares measured by the ILS over time. At the beginning of our sample period, Binance exhibited the highest price discovery share. However, this changed on March 22, 2023, when Binance terminated its aggressive marketing strategy that had previously allowed trading the BTC-USDT spot pair without fees. The initially high share of price discovery attributed to KuCoin in our sample period aligns with the launch of their own zero-fee trading program, introduced to celebrate the platform's fifth anniversary. This initiative, which significantly boosted Kucoin's market activity, is detailed in their official announcement[16]. However, after the program ended on November 16, 2022, KuCoin's contribution to price discovery sharply declined, nearing zero. Subsequently, most price discovery shifted to Binance, which continued offering its zero-fee program. The introduction of fees led to a sharp decline in trading volumes on Binance, which significantly impacted its price discovery share. Subsequently, Binance futures prices became the most informative, surpassing a 50% share. Coinbase spot prices are also important for price discovery, though their influence declines over time. Additionally, spot prices from Binance and OKX and futures prices from OKX contribute to price discovery.

[16] https://www.kucoin.com/announcement/en-0-fees-for-trading-btc-event-extension

Conversely, Kraken, Bitstamp, and Kucoin do not significantly contribute to price discovery in the Bitcoin spot market. These exchanges also exhibit the lowest trading volumes compared to the total sample. However, trading volumes for perpetual futures on unregulated exchanges are considerably higher, surpassing the spot volumes on the most prominent exchanges. We confirm prior research by Alexander et al. (2020) and Alexander and Heck (2020) that indicates that perpetual futures play a crucial role in price discovery.

	Regulated Spot	Unregulated Spot	Unregulated Futures	Platform	Market share
A: Full sample					
HIS	11.86	28.52	59.86	Regulated Spot	3.41
	(10.25)	(25.30)	(25.12)		
CS	16.03	27.13	56.84	Unregulated Spot	17.58
	(9.59)	(12.68)	(13.74)		
ILS	15.68	33.51	50.80	Unregulated Futures	79.01
	(16.70)	(35.39)	(30.94)		
B: Before March 22, 2023					
HIS	9.00	65.40	25.68	Regulated Spot	3.39
	(8.26)	(12.35)	(12.67)		
CS	16.66	37.45	45.89	Unregulated Spot	35.17
	(10.69)	(10.51)	(11.90)		
ILS	4.94	87.01	8.05	Unregulated Futures	61.44
	(6.00)	(9.12)	(8.34)		
C: After March 22, 2023					
HIS	12.97	14.15	73.18	Regulated Spot	3.41
	(10.72)	(9.67)	(13.48)		
CS	15.78	23.11	61.11	Unregulated Spot	10.73
	(9.11)	(11.07)	(11.93)		
ILS	19.87	12.67	67.47	Unregulated Futures	85.86
	(17.64)	(12.58)	(17.70)		

Table 3: This table displays the average price discovery share in percentages for the different price series in our sample. Standard deviations are shown in brackets below the averages. The top panel shows the numbers for the full sample from November 1, 2022 until March 31, 2024. The middle and bottom panels show the price discovery shares before and after introducing a trading fee on Binance on March 22, 2023.

Furthermore, to simplify the comparison of price discovery among the different groups of regulated and unregulated spot trading and perpetual futures trading, we can aggregate the price discovery shares to highlight which market segment is leading price discovery. Table 3 shows the grouped price discovery shares by market segment. For simplification, each exchange's individual price discovery shares are summed up[17]. From Table 3, we can make the following observations. First, the regulated spot segment is the least important for price discovery measured by either the HIS, CS, or ILS. However, it does contribute, on average, quite a high share (above 10%) compared to the relatively low volumes (only 3%) that are traded on the regulated spot platforms. Second, Panel B shows that the zero-fee programme on Binance boosts the market share of the unregulated spot market and leads to increased price discovery during that time. However, after the programme has been cancelled (see Panel C), these markets do not seem significantly more informative than the regulated markets despite having three times higher trading volumes. Finally, unregulated perpetual futures play an essential role in price discovery. Most volumes are traded in this market segment, and they have been leading price discovery for most of the sample period.

We also investigate whether price discovery exhibits intraday or weekday-specific patterns. The results, presented in Tables A.4 and A.5 in the Appendix, show that price discovery on Coinbase fluctuates throughout the day, with a pronounced increase during the US afternoon. This aligns with the timing of the New York Bitcoin fixing, which we explore further in Section 4.3. Beyond this intraday variation, price discovery remains relatively stable across hours. Regarding weekday effects, we observe a modest increase in price discovery on Coinbase and Kraken during weekends, accompanied by a slight decline on Binance Futures. However, no other significant differences emerge.

[17]The detailed results for every exchange are in the Appendix in Table A.3.

4.2 Relative Trading Volumes and Transaction Costs

Next, we investigate the relationship between price discovery (ILS[18]) and relative trading volumes, trading costs, and volatility. Specifically, we estimate the following regression model using hourly data:

$$ILS_{i,t} = \beta_0 + \beta_1 R_{i,t} + \beta_2 RTV_{i,t} + \beta_3 S_{i,t} + \beta_4 V_{i,t} + \rho ILS_{i,t-1} + \epsilon_{i,t}, \tag{9}$$

where $ILS_{i,t}$ is the information leadership measure, $R_{i,t}$ the hourly return, $RTV_{i,t}$ the relative trading volume, $S_{i,t}$ the effective spread and $V_{i,t}$ the hourly realized volatility for exchange i at time t. The relative trading volume is calculated as the traded volume divided by the sum of the total traded volume on all eight exchanges and instruments in hour t. The EDGE measure estimates the effective spread as explained in Section 2. We include one lag of the ILS measure to control for autocorrelation. As a robustness check, we include hourly dummies in the regression to account for intraday seasonality. The results remain broadly unchanged and are available from the authors upon request. As illustrated in Figure 1, we noticed a structural break on March 22, 2023, coinciding with Binance's discontinuation of their zero-fee trading program for the BTC-USDT spot pair. Therefore, we run separate regressions for the period with zero fees and afterwards. We run eight regressions one for every instrument and exchange. The results are depicted in Tables 4 and 5. Results for the entire sample period are provided in the Appendix in Table A.6.

[18]The main results also hold for the HIS and CS price discovery measures. Results are available from the authors upon request.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	-0.637**	0.330**	0.221***	0.022	53.401***	-0.174	-13.554***	-1.000***
	(0.318)	(0.166)	(0.053)	(0.173)	(3.796)	(0.110)	(1.617)	(0.321)
R (in %)	-0.000	-0.000	-0.001	-0.000	-0.005	0.002***	0.005	0.001
	(0.002)	(0.001)	(0.001)	(0.001)	(0.004)	(0.001)	(0.003)	(0.001)
RTV (in %)	1.583***	1.150**	0.706***	0.244	0.172***	0.578***	0.330***	0.155***
	(0.133)	(0.508)	(0.251)	(0.209)	(0.049)	(0.136)	(0.039)	(0.037)
S (in bps)	-0.108	-0.052***	-0.008	0.284***	-0.721**	0.301*	-0.014	0.035
	(0.098)	(0.018)	(0.009)	(0.101)	(0.340)	(0.157)	(0.646)	(0.099)
V (in %)	-0.008**	-0.002*	0.000	-0.002	-0.091***	-0.005	0.087***	0.016***
	(0.004)	(0.001)	(0.001)	(0.003)	(0.010)	(0.004)	(0.009)	(0.004)
AR(1)	0.145***	0.071	0.076**	0.788***	0.340***	0.015	0.129***	0.017
	(0.032)	(0.050)	(0.036)	(0.049)	(0.044)	(0.015)	(0.029)	(0.013)
Observations	3019	3019	3019	3019	3019	3019	3019	3019
Adjusted R^2	0.309	0.035	0.016	0.657	0.364	0.025	0.464	0.089

Table 4: This table shows the results of estimating Equation 9 on hourly data for eight different exchanges and instruments. The regression uses data from November 1, 2022, until March 22, 2023. ILS is the hourly information leadership measure in % computed as explained in Section 3 for a model containing traded prices from the eight instruments and exchanges displayed in this table. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized (annualized) volatility and AR(1) is one lag of the ILS. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	13.253***	1.170***	1.183***	-0.070	-0.561	-1.635***	7.374**	-3.006***
	(0.734)	(0.409)	(0.110)	(0.126)	(0.570)	(0.284)	(3.234)	(0.648)
R (in %)	-0.003	-0.001	-0.000	-0.001	-0.003	0.005*	0.001	0.000
	(0.003)	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.005)	(0.002)
RTV (in %)	1.591***	2.899***	0.974***	0.476**	0.590***	2.766***	0.538***	0.579***
	(0.155)	(0.725)	(0.315)	(0.187)	(0.066)	(0.133)	(0.053)	(0.031)
S (in bps)	-0.815***	-0.099	-0.060***	0.025	-0.184	-0.116	-3.269***	-0.223
	(0.206)	(0.069)	(0.014)	(0.059)	(0.518)	(0.142)	(1.007)	(0.169)
V (in %)	-0.166***	-0.027***	-0.008***	0.002	0.084***	0.031***	0.125***	0.021***
	(0.016)	(0.004)	(0.002)	(0.002)	(0.005)	(0.005)	(0.016)	(0.005)
AR(1)	0.330***	0.042	0.049***	0.015	0.090***	0.084***	0.211***	0.092***
	(0.017)	(0.028)	(0.018)	(0.014)	(0.019)	(0.018)	(0.014)	(0.013)
Observations	7749	7749	7749	7749	7749	7749	7749	7749
Adjusted R^2	0.246	0.082	0.017	0.003	0.080	0.127	0.096	0.094

Table 5: This table shows the results of estimating Equation 9 on hourly data for eight different exchanges and instruments. The regression uses data from March 22, 2023, until March 31, 2024. ILS is the hourly information leadership measure computed in % as explained in Section 3 for a model containing traded prices from the eight instruments and exchanges displayed in this table. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized (annualized) volatility and AR(1) is one lag of the ILS. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

Tables 4 and 5 present the relationship between trading volumes and price discovery, revealing that higher trading volumes, especially relative to other exchanges, enhance price discovery. This finding supports the theory proposed by Admati and Pfleiderer (1988), which suggests that higher trading volumes indicate informed trading. The impact is especially large in regulated spot markets. For instance, an increase of 1% in relative trading volume results in a 0.7% increase in price discovery for Bitstamp (the smaller impact), but it can also lead to a 2.9% increase in the post-fee period for Kraken (the largest effect). The results show that relative trading volumes especially lead to higher levels of price discovery. This is also in line with other investigations of the cryptocurrency market (among others Alexander and Heck, 2020; Entrop et al., 2020).

Furthermore, trading costs emerge as a crucial factor in price discovery. During Binance's zero-fee trading program, most price discovery occurred on Binance spot, as evidenced by a significant constant of more than 50%. However, even without the fee waiver, unregulated exchanges typically offer lower trading fees, particularly for takers, with perpetual futures contracts being cheaper than spot trades. Effective spreads were not a major consideration during this period, as the average effective spread on Binance (Table 2) was considerably lower than the trading fees on other venues (Table 1). This dynamic shifted following the introduction of trading fees on March 22, 2023. Post-fee implementation, effective spreads, as a measure of cost to trade on a specific exchange, became a significant determinant of price discovery, with a one basis point increase in effective spreads leading to between 0.8% (for Coinbase) and 3% (for Binance Futures) decrease in price discovery. Although this effect is significant, it is less pronounced compared to the impact of trading volumes. This indicates that while trading costs have become more relevant, their influence is still secondary to trading volume in driving price discovery. The finding that trading costs have a significant impact on price discovery is not surprising, and it has been found that best bid-offer spreads significantly affect price discovery (among others Frijns et al., 2015; Hauptfleisch et al., 2016).

Finally, the results indicate that higher realised volatility enhances price discovery in unregulated futures markets, consistent with the findings of Alexander and Heck (2020)

regarding the BitMEX perpetual futures market.

Interestingly, regulated spot markets, particularly Coinbase, are crucial for price discovery. Despite the Binance perpetual futures contract leading overall in Bitcoin price discovery, Coinbase's significance remains notable. Additionally, Coinbase consistently shows the highest significant constant across different regressions, suggesting the particular importance of regulated spot venues in the price discovery process.

4.3 Bitcoin New York Fixing

Previous results have shown that trades on regulated spot exchanges, such as Coinbase, can contain more information than on other exchanges. One potential explanation can be that traded prices from these exchanges are used as a reference to define prices for the broader market. For example, many popular ETFs on Bitcoin use the CME CF Bitcoin Reference Rate - New York Variant (BRRNY) as a reference price. This reference rate is the average volume-weighted median trade prices from eligible exchanges between 3 pm and 4 pm New York time. The list of eligible exchanges includes Coinbase, Kraken, and Bitstamp. Based on trading volumes, Coinbase is the most important contributor to the reference rate (see Figure 4). As the reference rate is widely used for pricing other financial products, price discovery on contributing venues might be increased during the fixing window. Figure 6 shows the average price discovery share on Coinbase per hour of the day relative to the daily average for the whole sample period. We notice that price discovery peaks at 7 and 8 pm UTC, which, depending on the daylight time change in New York, corresponds to the 3 pm until 4 pm New York fixing window.

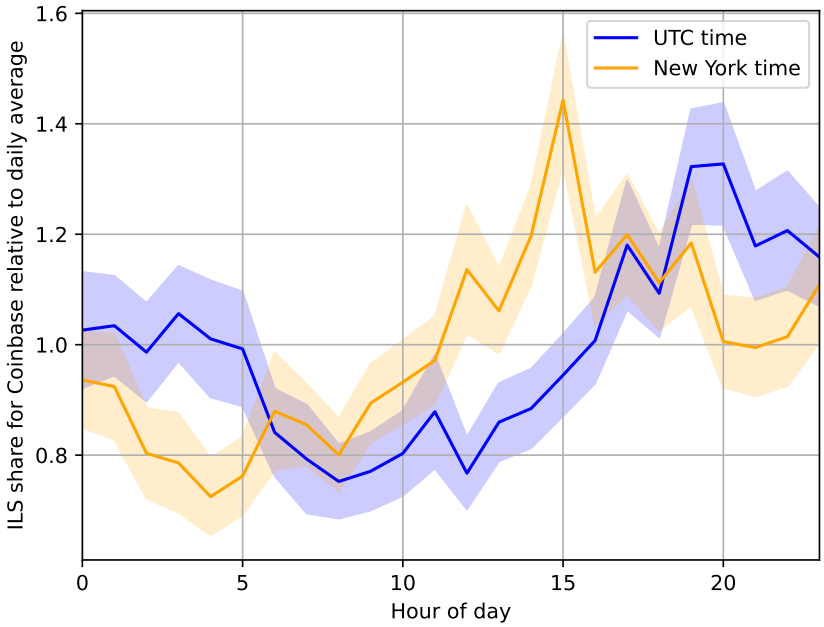

Figure 6: The average price discovery share for every hour of the day as measured by the ILS for trades on Coinbase. The price discovery share is normalised by its daily average to make results comparable. The shaded area are 95% confidence bounds. We show the results for UTC in blue and the results for New York time in orange. The average price discovery share peaks at 7 pm and 8 pm UTC, which, depending on the daylight time change, corresponds to the 3 pm to 4 pm New York fixing window. In New York time, it peaks always during the fixing window at 3 pm.

To investigate whether the fixing window significantly contributes to the importance of regulated spot venues for price discovery, we estimate the following model:

$$\begin{aligned} ILS_{i,t} = \beta_0 + \beta_1 RTV_{i,t} + \beta_2 S_{i,t} + \beta_3 R_{i,t} + \beta_4 V_{i,t} + \rho \Delta ILS_{i,t-1} \\ + D_t^{3pmNY} + D_t^{3pmNY} \times RTV_{i,t} + D_t^{3pmNY} \times S_{i,t} + \epsilon_{i,t}, \end{aligned} \tag{10}$$

where the variables are defined as in Equation 9, and we add a dummy variable D_t^{3pmNY}, which is true if the hourly observations occur between 3 pm and 4 pm New York time. We also include interaction terms with relative trading volume and effective spreads. The main results for the second period, after Binance reintroduced fees, are displayed in Table 6. The results for the whole period are shown in the Appendix in

Table A.7. The results during the zero-fee program are not significant[19].

The findings reveal that Coinbase exhibits an increase in price discovery of more than 10% during the New York fixing window. Market quality measures cannot explain this difference. Given the fact that Coinbase is the most important for defining the Bitcoin reference rate, the results show that price discovery strongly increases during the fixing window. On the other hand, we do not find an effect for Kraken and Bitstamp that also contributes to the fixing price. However, these venues are much less relevant in the calculation of the Bitcoin reference price, because of their low trading volume.

To assess whether the fixing window between 3 pm and 4 pm New York time has a causal effect on price discovery of the regulated spot exchanges, we use the daylight change in New York as a quasi-natural experiment. The daylight change in New York, i.e., Eastern Standard Time (EST), gets changed to Eastern Daylight Time (EDT) and the other way around, occurs twice a year. As a result, the time in New York changes relative to UTC by one hour four times in our sample. This presents an interesting quasi-natural experiment that can be exploited to identify whether price discovery on a specific venue depends on the 3 pm to 4 pm New York window. To do so, we first define the relative price discovery share as $RPD_t = \frac{PD_t}{\overline{PD}_d}$ where $\overline{PD}_d = \frac{1}{24}\sum_{t \in d} PD_t$, which is the price discovery share relative to the daily average value. Figure 6 shows the average RPD for Coinbase. We normalise the price discovery share in this way to keep it comparable throughout the whole sample period. We can select all price discovery estimates from our sample from 7 pm to 8 pm UTC and 8 pm to 9 pm UTC. Depending on the daylight change, we can use the UTC window corresponding to the 3 pm New York window as treatment and the other time window as a control group.

[19] see Table A.8 in the Appendix

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	20.006***	1.241***	1.253***	-0.083	-0.421	-1.731***	14.444***	-2.385***
	(0.966)	(0.389)	(0.121)	(0.134)	(0.599)	(0.301)	(4.086)	(0.712)
R	-0.005	-0.001	-0.001	-0.001	-0.003	0.005*	0.002	0.000
	(0.004)	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.005)	(0.002)
RTV	1.593***	2.993***	0.942***	0.504**	0.636***	3.000***	0.608***	0.601***
	(0.224)	(0.756)	(0.355)	(0.198)	(0.069)	(0.133)	(0.066)	(0.033)
S	-1.163***	-0.106	-0.059***	0.031	-0.273	-0.055	-3.528***	-0.238
	(0.242)	(0.073)	(0.014)	(0.062)	(0.538)	(0.150)	(1.141)	(0.180)
V	-0.197***	-0.027***	-0.008***	0.002	0.088***	0.032***	0.142***	0.022***
	(0.020)	(0.004)	(0.002)	(0.002)	(0.005)	(0.005)	(0.018)	(0.005)
D_t^{3pmNY}	10.486***	0.764	0.111	0.338**	-0.263	0.155	7.371	1.100
	(2.002)	(0.504)	(0.423)	(0.171)	(2.155)	(1.056)	(10.404)	(1.889)
$D_t^{3pmNY} \times RTV$	-1.035**	-2.120**	0.158	-0.536**	0.091	-0.168	-0.171	-0.131
	(0.422)	(0.876)	(0.805)	(0.220)	(0.263)	(0.574)	(0.160)	(0.098)
$D_t^{3pmNY} \times S$	-1.288	0.035	-0.048	-0.072	-0.823	-0.559	0.138	0.283
	(0.891)	(0.099)	(0.066)	(0.062)	(1.623)	(0.370)	(2.982)	(0.335)
Observations	7749	7749	7749	7749	7749	7749	7749	7749
Adjusted R^2	0.146	0.081	0.014	0.003	0.072	0.121	0.054	0.087

Table 6: This table shows the results of estimating Equation 10 on hourly data for eight different exchanges and instruments using data from March 22, 2023, until March 31, 2024. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized volatility and AR(1) is one lag of the ILS. D_t^{3pmNY} is a dummy term that is true if the hourly observations occurred between 3 pm and 4 pm New York time. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

Given that the daylight change occurs four times during our sample period, we can identify the causal effect using a difference-in-differences type of estimation. Specifically, we estimate the following regression:

$$RPD_t = \beta_0 + \beta_1 D_t^{7pmUTC} + \beta_2 D_t^{EDT} + \beta_3 D_t^{7pmUTC} \times D_t^{EDT} + \gamma X_t + \epsilon_t, \qquad (11)$$

where RPD_t contains all observations for 7 pm and 8 pm UTC. D_t^{7pmUTC} is a dummy is true if the time window is 7 pm to 8 pm UTC. Under EDT time, this corresponds to 3 pm to 4 pm New York time. D_t^{EDT} is a dummy that is true if EDT time is in place and $D_t^{7pmUTC} \times D_t^{EDT}$ is the interaction between the two dummies. X_t is a vector of control variables that contains returns, volatility, spread and relative traded volumes during the time window. Furthermore, in a second step, we can add a dummy variable D_t^{ETF} for the approval of ETFs by the SEC on January 8, 2024, to investigate whether this has led to an increasing effect on price discovery. Table 7 shows the regression results.

	(1)	(2)
D_t^{7pmUTC}	-0.105	-0.122
	(0.137)	(0.137)
D_t^{EDT}	-0.201	-0.228*
	(0.128)	(0.129)
$D_t^{7pmUTC} \times D_t^{EDT}$	0.346**	0.328*
	(0.170)	(0.170)
$D_t^{7pmUTC} \times D_t^{EDT} \times D_t^{ETF}$		0.414*
		(0.238)
Controls	Yes	Yes
Observations	682	682
Adjusted R^2	0.178	0.180

Table 7: The results of estimating Equation 11 on all observations after the Binance zero-fee period that covers the two hours between 7 pm and 9 pm UTC. Return, Volatility, RTV, and Spread are used as control variables. D_t^{7pmUTC} is a dummy that is true for observations between 7 pm UTC and 8 pm UTC. D_t^{EDT} is true if the observation was during EDT and false otherwise. D_t^{ETF} is true after the approval of Bitcoin ETFs by the SEC on January 8, 2024. We use heteroscedasticity and autocorrelation consistent standard errors and *, ** and *** denotes significance at the 10%, 5%, and 1% level.

The results show a causal and significantly positive effect of the 3 pm to 4 pm New York time fixing window on the relative price discovery on Coinbase. Controlling for

variables such as return, volatility, relative traded volume and spreads, price discovery is still 34.6% higher during that time window than during the rest of the day. The value is economically significant. Calculated based on the average price discovery share on Coinbase during our sample (which is 16.82%; see Table A.3 in the Appendix), this accounts for an increase of around 5.8% to overall price discovery. This means that the fixing window is important for price discovery and increases the importance of regulated exchanges in the sample. Moreover, we also investigate whether the effect, with the approval of the Bitcoin ETFs, has been increasing as the New York fixing rate is a benchmark in most Bitcoin ETFs. Thus, knowing whether regulated or unregulated exchanges determine prices during that time is essential. Since the approval of Bitcoin ETFs, the relative importance during that time window has increased. However, as our sample only covers about three months since the introduction of Bitcoin ETFs this effect might further increase over time. It will be interesting for future research to study whether the approval of Bitcoin ETFs will lead to changes in where prices are determined.

Again, we find no significant effect for Kraken and Bitstamp, both exchanges also used in the fixing. However, this might not be surprising given that Coinbase's trading volumes are much higher during the fixing window. For the unregulated exchanges, we find no significant effect.

4.4 Robustness

Tardis provides us with two types of timestamps: the exchange-provided timestamps and those recorded when Tardis captures the data on its servers. For our analysis, we rely on the exchange timestamps, which reflect the exact moment a trade is matched within the exchange's matching engine and when the information is publicly disseminated. This timestamp is also received by other market participants consuming the data, making it more relevant for our study of price discovery. Additionally, using Tardis' timestamps would introduce additional latency, as their servers are not co-located with the exchanges, potentially adding delays to the recorded time of each trade. However, one limitation of relying on exchange timestamps is that the clocks of different exchanges are not syn-

chronised, potentially introducing inaccuracies. To ensure the robustness of our analysis, we recalculated all price discovery metrics for one month using Tardis' timestamps. As shown in Table A.12 in the Appendix, the differences between the two timestamps are small, supporting our decision to use the exchange timestamps for the complete analysis.

In addition, we run a series of robustness checks on the regressions in Section 4.3. First, we add placebo dummies and investigate whether moving the daylight time change changes the validity of the results. We move the daylight change dates by three months and repeat the regression. This can be found in the Appendix in Table A.9. Second, we use the alternative time dummies at 5 pm and 9 pm UTC to check if other time windows have a significant effect. Table A.10 in the Appendix displays the results. Last, we investigate potential effects for other exchanges and instruments during the same time window. The results are in the Appendix in Table A.11. The effects of all robustness checks do not produce significant results, highlighting the validity of our causal finding that price discovery on the most important regulated spot exchange significantly increased during the New York fixing window.

Our main results, consistent with previous studies such as Baur and Dimpfl (2019), Alexander et al. (2020), and Alexander and Heck (2020), are based on traded prices. This raises the question of whether the price discovery shares are biased towards exchanges with higher trading volumes due to more frequent price updates. To address this, we also compute price discovery shares over a short interval using quoted mid-prices from the selected exchanges and compare them with our results based on traded prices.[20] Overall, our previously stated results also hold based on quoted prices. Binance continues to dominate in price discovery. Moreover, lower transaction costs and higher trading volumes remain critical factors in explaining price discovery.

[20]Results are not reported for brevity but are available from the authors upon request.

5 Conclusion

This study sheds light on the dynamics of price discovery within the fragmented and largely unregulated Bitcoin market. Using high-frequency data from major exchanges and the novel model by Hasbrouck (2021), our analysis reveals that exchanges facilitating trading against stablecoins like USDT and offering high-leverage perpetual futures play a dominant role in determining the price of Bitcoin. These platforms, driven by high trading volumes and lower transaction costs, have become the primary venues for price formation. Furthermore, our findings underscore the significant impact of changes on trading platforms, such as zero-fee trading and the widespread availability of perpetual futures contracts. These factors can significantly enhance the informativeness of prices on these exchanges. On the one hand, the concentration of price discovery within exchanges that have faced allegations of fraud and price manipulation raises serious concerns about market integrity and the effectiveness of regulatory oversight. On the other hand, we find that exchanges contributing to the New York 4 pm fixing are relatively more important for price discovery during that window. However, USDT-based exchanges continue to dominate price discovery, making them a crucial component of the fixing price. Given the limited regulatory oversight in these markets, our results highlight important challenges for financial regulators.

As Bitcoin becomes more popular among investors from the traditional financial systems, the insights from this study have important implications for both market participants and regulators. Prices of financial products, such as Bitcoin spot ETFs, are primarily determined by exchanges and instruments prone to price manipulation. Thus, ensuring market integrity and investor protection in this rapidly evolving asset class is crucial as cryptocurrencies continue to gain prominence.

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A Additional Figures and Tables

r	Lmax	Trace	Critical Values
7	4.71	4.71	11.65
6	34.75	39.46	23.52
5	64.34	103.80	37.22
4	210.13	313.93	55.43
3	268.72	582.65	78.87
2	330.64	913.30	104.20
1	518.90	1432.20	136.06
0	1512.16	2944.37	168.92

Table A.1: This table shows as an example the results of the Johansen (1991) cointegration test for all spot and futures trades on different exchanges between 00:00 and 01:00 UTC on December 2, 2022. The rank is denoted with r. The Trace statistic shows the test outcome for the null hypothesis H_0: $\text{rank}(\Pi) = r$, against the alternative hypothesis H_1 suggesting that $\text{rank}(\Pi) > r$. Additionally, Lmax indicates the maximum eigenvalue test result for H_0: $\text{rank}(\Pi)$ versus H_1: $\text{rank}(\Pi) = r + 1$. Finally, there are the critical values. For this example, we can reject the null hypotheses that exactly $r = 0, 1, 2, 3, 4, 5, 6$ cointegrating relationships exist.

	Mean	Median	Std Dev	Min	Max
Coinbase Spot					
Effective	1.462	1.178	0.953	0.627	6.098
Abdi-Ranaldo	6.167	3.957	7.114	0.554	104.349
EDGE	1.860	1.338	1.747	0.032	14.990
Kraken Spot					
Effective	3.582	2.589	3.052	0.563	16.124
Abdi-Ranaldo	5.497	3.395	6.376	0.000	51.761
EDGE	2.663	1.806	2.669	0.060	22.178
Bitstamp Spot					
Effective	3.732	2.372	3.993	0.893	24.053
Abdi-Ranaldo	5.778	3.724	6.559	0.136	58.698
EDGE	3.696	2.590	4.070	0.074	44.560
Kucoin Spot					
Effective	1.241	0.957	1.039	0.284	6.577
Abdi-Ranaldo	5.939	3.755	6.514	0.436	70.535
EDGE	1.325	0.882	1.411	0.016	14.270
Binance Spot					
Effective	1.799	1.409	1.223	0.704	7.843
Abdi-Ranaldo	6.039	3.912	6.431	0.053	70.509
EDGE	1.318	0.943	1.324	0.056	14.300
Binance Futures					
Effective	1.992	1.513	1.553	0.459	8.730
Abdi-Ranaldo	6.107	3.963	6.422	0.314	63.502
EDGE	0.544	0.369	0.827	0.021	14.081
OKX Spot					
Effective	2.455	1.583	3.611	0.541	29.283
Abdi-Ranaldo	6.019	3.945	6.435	0.438	72.123
EDGE	1.618	1.274	1.505	0.000	16.138
OKX Futures					
Effective	1.842	1.321	1.946	0.335	16.328
Abdi-Ranaldo	6.060	3.986	6.443	0.371	69.968
EDGE	1.268	0.967	1.131	0.055	9.253

Table A.2: We calculate the effective spread in basis points for every hour from November 2022. We then compute the spread estimator with the methods proposed by Abdi and Ranaldo (2017) and Ardia et al. (2024) (EDGE) based on data sampled at a five-minute frequency and aggregated per hour. This Table presents summary statistics for the effective spreads and the estimated spreads per exchange.

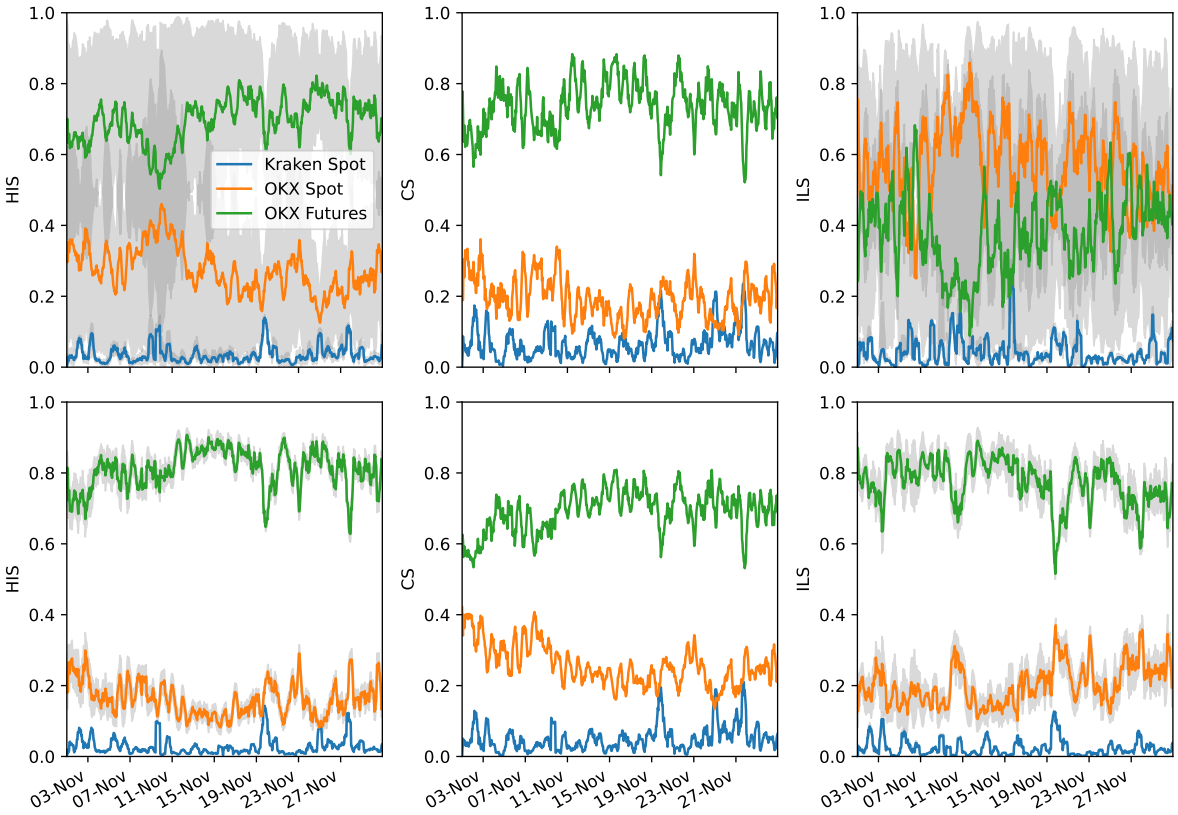

Figure A.1: This figure shows the estimates for different measures of price discovery in a system of three-time series from different exchanges and instruments. The upper and lower bounds of the HIS are reflected by the area shaded in grey. The upper row depicts estimates based on a one-second data aggregation, and the lower row displays the estimates at a frequency of ten milliseconds.

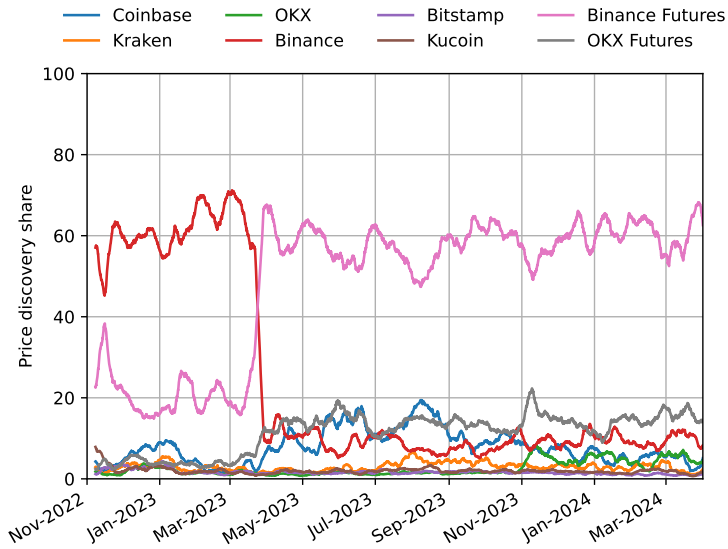

Figure A.2: This figure shows the 7-day moving average of the price discovery share computed based on the Hasbrouck information share. The model is estimated based on traded prices from eight different exchanges and instruments between November 1, 2022 and March 31, 2024.

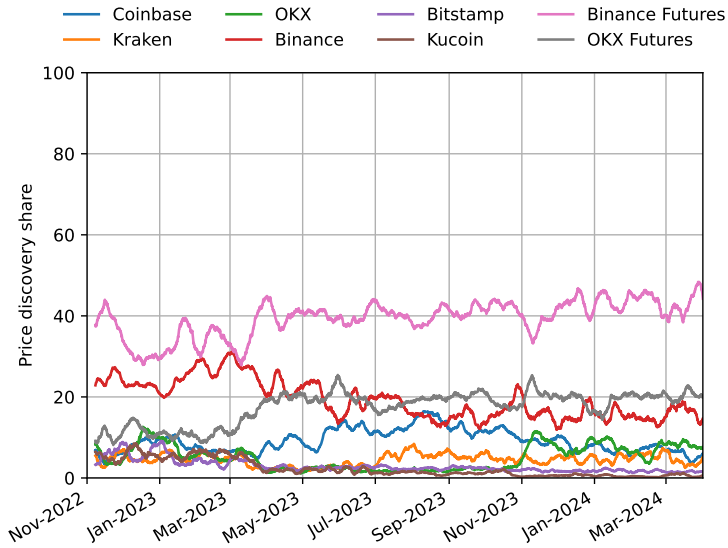

Figure A.3: This figure shows the 7-day moving average of the price discovery share computed based on the component share. The model is estimated based on traded prices from eight different exchanges and instruments between November 1, 2022 and March 31, 2024.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
A: Full sample								
HIS	7.43	2.83	1.59	2.00	23.91	2.62	48.58	11.28
	(8.03)	(5.32)	(2.72)	(2.93)	(25.14)	(4.10)	(21.99)	(10.12)
CS	8.76	4.40	2.87	2.38	19.58	5.17	39.50	17.35
	(5.43)	(7.09)	(4.65)	(5.07)	(9.69)	(6.19)	(11.06)	(9.63)
ILS	13.20	1.53	0.95	0.91	28.85	3.76	41.94	8.86
	(15.25)	(4.54)	(2.54)	(4.31)	(35.75)	(7.76)	(26.84)	(8.83)
B: Before March 22, 2023								
HIS	4.30	2.68	2.03	2.44	61.22	1.74	21.65	4.04
	(5.05)	(5.00)	(3.57)	(3.54)	(12.61)	(2.95)	(11.79)	(5.49)
CS	7.09	4.78	4.79	5.63	25.40	6.42	34.40	11.49
	(4.63)	(7.46)	(6.70)	(7.27)	(6.18)	(7.47)	(11.21)	(9.17)
ILS	3.93	0.59	0.42	2.30	84.13	0.58	6.84	1.21
	(5.30)	(1.83)	(1.09)	(5.96)	(11.26)	(2.66)	(7.14)	(2.81)
C: After March 22, 2023								
HIS	8.66	2.89	1.42	1.82	9.36	2.97	59.08	14.10
	(8.63)	(5.43)	(2.28)	(2.63)	(7.90)	(4.42)	(14.99)	(10.10)
CS	9.41	4.25	2.12	1.11	17.31	4.69	41.48	19.63
	(5.57)	(6.93)	(3.25)	(3.06)	(9.86)	(5.54)	(10.34)	(8.81)
ILS	16.82	1.90	1.15	0.37	7.31	4.99	55.63	11.84
	(16.30)	(5.18)	(2.89)	(3.30)	(8.48)	(8.68)	(17.71)	(8.58)

Table A.3: This table displays the average price discovery share in percentages for the different price series in our sample. HIS stands for the Hasbrouck information share, CS stands for the Gonzalo Granger component share, and ILS stands for the information leadership share. Standard deviations are shown in brackets below the averages. The top panel shows the numbers for the full sample from November 1, 2022 until March 31, 2024. The middle and bottom panels show the price discovery shares before and after introducing a trading fee on Binance on March 22, 2023.

Hour	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
0	0.129	0.016	0.011	0.009	0.294	0.032	0.417	0.092
1	0.137	0.013	0.010	0.008	0.293	0.039	0.403	0.097
2	0.137	0.026*	0.013	0.008	0.300	0.037	0.382*	0.097
3	0.145	0.016	0.011	0.008	0.297	0.047	0.380*	0.096
4	0.134	0.019	0.011	0.009	0.296	0.044	0.387	0.100
5	0.124	0.014	0.009	0.009	0.300	0.043	0.405	0.096
6	0.110*	0.018	0.010	0.012	0.310	0.040	0.409	0.093
7	0.102*	0.016	0.011	0.010	0.305	0.058*	0.398	0.101
8	0.102*	0.012	0.010	0.008	0.307	0.038	0.429	0.094
9	0.113*	0.014	0.012	0.009	0.286	0.038	0.428	0.101
10	0.112*	0.016	0.009	0.009	0.300	0.035	0.434	0.086
11	0.119	0.018	0.011	0.008	0.285	0.037	0.436	0.085
12	0.104*	0.010*	0.008	0.008	0.277	0.040	0.454*	0.100
13	0.110*	0.012	0.010	0.010	0.276	0.043	0.447	0.093
14	0.119	0.010*	0.007*	0.007	0.276	0.028*	0.461*	0.092
15	0.121	0.013	0.006*	0.009	0.278	0.040	0.441	0.092
16	0.135	0.012	0.007*	0.007	0.274	0.031*	0.447	0.087
17	0.147	0.015	0.009	0.010	0.278	0.029*	0.432	0.080
18	0.146	0.013	0.006*	0.012	0.280	0.031	0.433	0.078*
19	0.172*	0.013	0.010	0.009	0.280	0.033	0.413	0.068*
20	0.162*	0.014	0.009	0.010	0.285	0.035	0.417	0.067*
21	0.157*	0.015	0.009	0.010	0.283	0.033	0.410	0.082
22	0.162*	0.022	0.012	0.009	0.274	0.041	0.402	0.079
23	0.153*	0.021	0.010	0.009	0.302	0.037	0.389	0.079

Table A.4: This table shows the hourly average price discovery share based on the ILS per exchange and instrument over our whole sample. The hours indicate the starting hour of the time window in UTC time. We perform a t-test on the hypothesis that the mean in a specific hour equals the overall mean for that exchange. For values marked with a star (*) we can reject that hypothesis at the 1% level.

Day	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Mon	0.123	0.014	0.010	0.008	0.287	0.040	0.424	0.095
Tue	0.124	0.014	0.009	0.009	0.299	0.039	0.421	0.085
Wed	0.116*	0.014	0.009	0.007	0.287	0.034	0.442*	0.089
Thu	0.130	0.015	0.009	0.008	0.284	0.034	0.429	0.090
Fri	0.139	0.013	0.009	0.008	0.284	0.040	0.421	0.086
Sat	0.156*	0.021*	0.011	0.010	0.289	0.037	0.393*	0.082
Sun	0.143	0.018	0.011	0.012	0.290	0.039	0.395*	0.093

Table A.5: This table shows the average price discovery per day of the week share based on the ILS per exchange and instrument over our whole sample. We perform a t-test on the hypothesis that the mean on a specific weekday equals the overall mean for that exchange. For values marked with a star (*) we can reject that hypothesis at the 1% level.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	8.000***	0.700**	0.773***	-0.283	-6.643***	-1.598***	-48.117***	-5.636***
	(0.615)	(0.311)	(0.087)	(0.283)	(0.525)	(0.185)	(1.652)	(0.240)
R	-0.001	-0.001	-0.000	-0.001	-0.004	0.004**	0.001	0.000
	(0.002)	(0.001)	(0.000)	(0.001)	(0.003)	(0.002)	(0.003)	(0.001)
RTV	1.186***	2.860***	1.155***	0.640*	1.122***	2.840***	1.114***	0.684***
	(0.132)	(0.621)	(0.266)	(0.382)	(0.070)	(0.114)	(0.035)	(0.018)
S	-0.686***	-0.098**	-0.035***	0.119*	2.334***	-0.037	-3.258***	-0.263***
	(0.144)	(0.043)	(0.010)	(0.062)	(0.469)	(0.111)	(0.589)	(0.093)
V	-0.091***	-0.016***	-0.005***	0.005*	0.037***	0.013***	0.067***	0.012***
	(0.009)	(0.002)	(0.001)	(0.003)	(0.007)	(0.004)	(0.012)	(0.003)
AR(1)	0.439***	0.054**	0.066***	0.459***	0.570***	0.094***	0.544***	0.113***
	(0.016)	(0.026)	(0.017)	(0.080)	(0.025)	(0.017)	(0.014)	(0.013)
Observations	10769	10769	10769	10769	10769	10769	10769	10769
Adjusted R^2	0.276	0.084	0.019	0.245	0.924	0.170	0.625	0.348

Table A.6: This table shows the results of estimating Equation 9 on hourly data for eight different exchanges and instruments. The regression uses data from November 1, 2022 until March 31, 2024. ILS is the hourly information leadership measure in % computed as explained in Section 3 for a model containing traded prices from the eight instruments and exchanges displayed in this table. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized (annualized) volatility and AR(1) is one lag of the ILS. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	15.428***	0.765**	0.835***	-0.440	-14.444***	-1.751***	-96.563***	-6.144***
	(0.992)	(0.301)	(0.094)	(0.492)	(0.567)	(0.196)	(3.337)	(0.255)
R	-0.002	-0.001	-0.000	-0.002**	-0.001	0.004**	0.003	0.000
	(0.003)	(0.001)	(0.000)	(0.001)	(0.004)	(0.002)	(0.004)	(0.001)
RTV	1.017***	2.978***	1.190***	1.122*	2.533***	3.140***	2.323***	0.765***
	(0.232)	(0.654)	(0.294)	(0.665)	(0.045)	(0.112)	(0.058)	(0.016)
S	-1.215***	-0.107**	-0.036***	0.190**	7.220***	-0.013	-6.018***	-0.304***
	(0.178)	(0.045)	(0.010)	(0.096)	(1.037)	(0.117)	(0.789)	(0.095)
V	-0.112***	-0.017***	-0.005***	0.010**	0.068***	0.013***	0.084***	0.011***
	(0.012)	(0.002)	(0.001)	(0.005)	(0.011)	(0.004)	(0.019)	(0.003)
D_t^{3pmNY}	6.544***	0.655*	0.017	-1.416**	0.460	0.675	22.529***	1.672***
	(1.703)	(0.390)	(0.334)	(0.698)	(0.837)	(0.668)	(5.874)	(0.573)
$D_t^{3pmNY} \times RTV$	-0.102	-1.975***	-0.088	1.490*	-0.003	-0.536	-0.426***	-0.121***
	(0.365)	(0.752)	(0.716)	(0.879)	(0.080)	(0.438)	(0.096)	(0.039)
$D_t^{3pmNY} \times S$	-1.269**	0.020	0.002	-0.032	-1.009	-0.298	-2.122	0.082
	(0.615)	(0.062)	(0.040)	(0.147)	(1.210)	(0.216)	(2.989)	(0.202)
Observations	10769	10769	10769	10769	10769	10769	10769	10769
Adjusted R^2	0.092	0.082	0.015	0.043	0.883	0.162	0.446	0.339

Table A.7: This table shows the results of estimating Equation 10 on hourly data for eight different exchanges and instruments. The regression uses data from November 1, 2022 until March 31, 2024. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized volatility and AR(1) is one lag of the ILS. D_t^{3pmNY} is a dummy term that is true if the hourly observations occurred between 3 pm and 4 pm New York time. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	Coinbase Spot	Kraken Spot	Bitstamp Spot	Kucoin Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures
Constant	-0.490	0.359**	0.235***	0.111	80.732***	-0.210	-14.589***	-0.967***
	(0.316)	(0.148)	(0.058)	(0.775)	(2.249)	(0.131)	(1.774)	(0.344)
R	0.000	-0.000	-0.001	-0.003**	-0.004	0.002***	0.004	0.001
	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.001)	(0.003)	(0.001)
RTV	1.730***	1.221**	0.806***	1.150	0.242***	0.633***	0.364***	0.152***
	(0.118)	(0.559)	(0.274)	(0.882)	(0.062)	(0.177)	(0.043)	(0.040)
S	-0.138	-0.053***	-0.008	0.910***	-1.100***	0.306*	0.011	0.049
	(0.100)	(0.019)	(0.010)	(0.279)	(0.372)	(0.164)	(0.692)	(0.104)
V	-0.007**	-0.002*	0.000	0.007	-0.111***	-0.005	0.092***	0.017***
	(0.003)	(0.001)	(0.001)	(0.009)	(0.013)	(0.004)	(0.010)	(0.004)
D_t^{3pmNY}	-1.433	0.335	0.081	-4.092**	2.650	0.357*	-5.787	-0.391
	(1.551)	(0.262)	(0.233)	(1.866)	(5.697)	(0.199)	(6.425)	(0.579)
$D_t^{3pmNY} \times RTV$	0.592	-0.930	-0.666	3.574*	-0.137	-0.451*	0.108	0.052
	(0.442)	(0.680)	(0.467)	(1.958)	(0.145)	(0.254)	(0.137)	(0.078)
$D_t^{3pmNY} \times S$	-0.067	-0.028	0.022	0.513	0.655	-0.061	-1.588	-0.227
	(0.269)	(0.040)	(0.038)	(0.511)	(1.260)	(0.127)	(2.173)	(0.212)
Observations	3019	3019	3019	3019	3019	3019	3019	3019
Adjusted R^2	0.294	0.031	0.011	0.097	0.262	0.025	0.450	0.089

Table A.8: This table shows the results of estimating Equation 10 on hourly data for eight different exchanges and instruments. The regression uses data from November 1, 2022 until March 22, 2023. R is the hourly return, RTV the relative trading volume, S the effective spread, V the realized volatility and AR(1) is one lag of the ILS. D_t^{3pmNY} is a dummy term that is true if the hourly observations occurred between 3 pm and 4 pm New York time. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	(1)
D_t^{7pmUTC}	0.291**
	(0.125)
$D_t^{EDT,Placebo}$	0.068
	(0.114)
$D_t^{7pmUTC} \times D_t^{EDT,Placebo}$	-0.278*
	(0.160)
Controls	Yes
Observations	686
Adjusted R^2	0.177

Table A.9: The results of estimating Equation 11 on all observations that cover the two hours between 7 pm and 9 pm UTC. Return, Volatility, RTV and Spread are used as control variables. D_t^{7pmUTC} is a dummy that is true if the price discovery share was estimated based on traded prices between 7 pm UTC and 8 pm UTC and false if between 8 pm UTC and 9 pm UTC. $D_t^{EDT,Placebo}$ is true if the observation was during Eastern Daylight Time and false if during Eastern Standard Time but shifted by three months to serve as a placebo test. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	(1)	(2)
D_t^{5pmUTC}	0.076	
	(0.107)	
D_t^{9pmUTC}		-0.060
		(0.131)
D_t^{EDT}	0.048	0.151
	(0.096)	(0.115)
$D_t^{5pmUTC} \times D_t^{EDT}$	-0.051	
	(0.132)	
$D_t^{9pmUTC} \times D_t^{EDT}$		0.037
		(0.162)
Controls	Yes	Yes
Observations	693	660
Adjusted R^2	0.118	0.075

Table A.10: The results of estimating Equation 11 on all observations that cover the two hours between 5 pm and 7 pm UTC (specification 1) and the two hours between 9 pm and 11 pm UTC (specification 2). Return, Volatility, RTV and Spread are used as control variables. D_t^{5pmUTC} and D_t^{9pmUTC} are dummies that are true if the price discovery share was estimated based on traded prices between the respective time window and false otherwise. $D_t^{EDT,Placebo}$ is true if the observation was during Eastern Daylight Time and false if it was during Eastern Standard Time. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	Kraken Spot	Binance Spot	OKX Spot	Binance Futures	OKX Futures	Kucoin Spot	Bitstamp Spot
D_t^{7pmUTC}	-0.045	0.080	0.161	0.014	-0.021	-0.134	0.180
	(0.210)	(0.128)	(0.206)	(0.039)	(0.065)	(0.388)	(0.296)
D_t^{EDT}	0.170	0.086	0.496**	-0.015	0.067	0.357	0.335
	(0.187)	(0.116)	(0.210)	(0.039)	(0.058)	(0.347)	(0.267)
$D_t^{7pmUTC} \times D_t^{EDT}$	0.030	-0.186	-0.354	-0.057	-0.009	-0.000	-0.003
	(0.261)	(0.160)	(0.256)	(0.049)	(0.080)	(0.483)	(0.368)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	682	686	686	686	686	675	686
Adjusted R^2	0.037	0.025	0.034	0.051	0.078	-0.001	0.019

Table A.11: The results of estimating Equation 11 on all observations that cover the two hours between 7 pm and 9 pm UTC. Return, Volatility, RTV and Spread are used as control variables. D_t^{7pmUTC} is a dummy that is true if the price discovery share was estimated based on traded prices between 7 pm UTC and 8 pm UTC and false if between 8 pm UTC and 9 pm UTC. D_t^{EDT} is true if the observation was during Eastern Daylight Time and false if during Eastern Standard Time. We use Newey and West (1987) heteroscedasticity and autocorrelation consistent standard errors with 24 hourly lags. *, ** and *** denote the coefficient estimates are significant at 10%, 5%, and 1% significance levels, respectively.

	CS		HIS		ILS	
	Tardis	Exchange	Tardis	Exchange	Tardis	Exchange
Coinbase Spot	0.077	0.092	0.064	0.076	0.129	0.133
Kraken Spot	0.068	0.059	0.044	0.043	0.020	0.028
Bitstamp Spot	0.019	0.019	0.016	0.016	0.014	0.015
Kucoin Spot	0.009	0.006	0.019	0.020	0.002	0.003
OKX Futures	0.217	0.209	0.195	0.166	0.180	0.142
OKX Spot	0.117	0.093	0.081	0.060	0.096	0.082
Binance Futures	0.357	0.377	0.500	0.538	0.484	0.519
Binance Spot	0.137	0.144	0.086	0.084	0.075	0.078

Table A.12: This table presents the average price discovery shares for November 2023, comparing three different metrics: Hasbrouck Information Share (HIS), Gonzalo Granger Component Share (CS), and Information Leadership Share (ILS). The "Tardis" column reflects price discovery shares calculated using timestamps provided by Tardis, while the "Exchange" column shows shares based on timestamps from the exchanges themselves.